6.5.3 Chemical soil degradation

6.5.3.1 Soil acidity

6.5.3.1.1 Development of soil acidity

Soils become acidic when a considerable portion of the exchangeable cations are hydrogen and hydrated aluminium. High acidity adversely affects the productivity of soil which results in stunted growth and reduced crop yields.

Reflection

What are the causes of soil acidity?

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The major causes of soil acidity include;

1. Acidic parent material

Some soil parent materials are acid. The differences in chemical composition of parent materials affect the acidity of soils. Therefore, soils that develop from granite materials are likely to be more acidic than soils developed from limestone.

2. Rainfall and leaching

As water containing various weak acids moves through the soil, some of the hydrogen ions from the weak acids replace adsorbed exchangeable cations. The exchangeable cations are leached into deeper soil profiles or ground water. Excessive rainfall is an effective agent for removing basic cations if a lot of water moves through the soil rapidly.

3. Organic matter decomposition and soil respiration

The carbon dioxide from decomposing organic matter and root respiration dissolves in water to form weak carbonic acid as shown in the equation below;

$CO_2 + HOH \rightarrow H_2CO_3 \rightarrow HCO_3^- + H^+$

Several organic acids are also produced during decomposition of organic matter.

4. Nitrification of ammonium

The use of fertilizers, especially those supplying nitrogen causes of soil acidity. Soil acidity is produced when ammonium containing fertilizer are transformed to nitrate. The fertilizers containing ammonium are oxidized by bacteria to form nitrate and hydrogen ions.

$NH_4^+ + 2O_2 \rightarrow NO_3^- + H_2O + 2H^+$

5. Acid rain

Carbon dioxide in the atmosphere reacts with rain to form carbonic acid which gives natural precipitation a pH of about 5.6. Industrial gases contain nitrogen and sulfur that result in the formation of acids hence causing acid rain. Acid rain falls when sulphur and nitrogen oxides are converted to sulphuric acid and nitric acid through oxidation and dissolving in rain drops.

6. Crop removal

Loss of basic cations from soils by cropping is one of the reasons why they become acidic (Tisdale 1985). Harvesting of crops makes soil acidic through uptake of cations such as calcium, magnesium and potassium, for their nutrition. When a cation is absorbed by a plant, a positively charged hydrogen ion is excreted into the soil to maintain electrical balance hence reducing the pH of soils.

An elaboration on causes of soil acidity is provided by Zhang (2013) in cause and effects of soil acidity.

6.5.3.1.2 Effects of soil acidity on soil properties

Soil acidity affects:

1. Plant nutrient availability which results in nutrient deficiencies due to:
   - Leaching of basic cations such as, potassium, calcium and magnesium
   - Reduced availability of phosphorus by fixation and precipitation
   - Molybdenum fixation into insoluble oxides of aluminium and iron

2. Soil levels of elements potentially toxic to plants. Elements such as aluminium, manganese and some micronutrients become very soluble resulting in solution concentration toxic to plants. In most acid soils, pH < 4, free strong acids exist. Pyrite (FeS_2) occurs in some soils and the sulphur is oxidized to sulphuric acid. The soils containing strong acids may produce toxicity of both aluminium and iron. The toxic elements are the major causes for crop failure in acid soils.

3. Soil microbial activity hence reducing the rate nitrogen fixation, nitrification, decomposition of organic matter and mineralization of nutrient elements. For example, the activity of bacteria and actinomycetes falls at pH values below 5.5.

6.5.3.1.3 Management of soil acidity

Alleviation of soil acidity includes growing acid tolerant crop varieties and application of lime and organic materials. Lime is a suitable material added to raise soil pH. In most cases, liming to pH 6.5 is recommended. However, certain specific crops such as watermelon and pineapples are better adjusted to lower pH values. In addition, some plants grow better if soils are moderately acid rather than neutral or basic pH. Therefore, soils can be acidified using materials such as sulphur, sulphuric acid and iron sulphate so as to aid plant availability of micronutrients and to lessen infections from pathogens (e.g. rots)

The principle of liming a soil is to apply a material which will react with the soil, removing hydrogen ions and replacing them with calcium or magnesium. The three major types of liming materials are the calcium and magnesium compounds- carbonates, oxides hydroxides. Calcium carbonate is one of the mostly used materials. The adsorbed hydrogen ions are replaced by calcium ions from lime. The hydrogen ions released are then neutralized by the carbonates and hydroxides as shown below:

Adsorbed $2H^+ + CaCO_3$ ► Adsorbed $Ca + H_2O + CO_2$

Calcium oxide and hydroxide behave in a similar manner

$CaO + H_2O$ ► $Ca(OH)_2$

Adsorbed $2H^+ + Ca(OH)_2$ ► Adsorbed $Ca + 2H_2O$

Calcium hydroxide and Calcium carbonate react with carbonic acid according to the following reactions:

$Ca(OH)_2 + 2H_2CO_3$ ► $Ca(HCO_3)_2 + 2H_2O$

$CaCO_3 + H_2CO_3$ ► $Ca(HCO_3)_2$ ---- calcium bicarbonate/calcium hydrogen carbonate

Adsorbed $2H^+ + Ca(HCO_3)_2$ ► Adsorbed $Ca + 2H_2O + 2CO_2$

Liming materials precipitate Al^{3+} into insoluble compounds inhibiting it from releasing H^+

6.5.3.2 Soil salinity and sodicity

6.5.3.2.1 Development of soil salinity and sodicity

Salted soils are unproductive for most crops unless harmful salts are removed. Soluble salts are inorganic compounds more soluble than gypsum which has a solubility of 0.214 g per 100 cm^3 at 0 °C. The salts are mainly chlorides, carbonates and sulphates of calcium, magnesium, potassium and sodium. Salted soils include saline soils, sodic soils and saline-sodic soils.

Saline soils contain a significant amount of salts to impair plant growth. The electrical conductivity gives a measure of salinity. Saline soils have an electrical conductivity of the saturated-soil extract of 4 or more mS/cm. Sodic soils have Sodium Absorption Ratio (SAR) of the saturation extract greater than 13 or Exchangeable Sodium Percentage (ESP) of 15 or more. Saline-sodic soils are characterized by high levels of salts and SAR over 13 or 15 percent or more of exchangeable sodium.

Table 9: Classification of salt affected soils

Name of soil	EC (mS/cm)	Soil pH	SAR $Na^+/\sqrt{(Ca^{2+}+Mg^{2+})/2}$	ESP
Normal (Non saline –Non Sodic)	< 4.0	-----	< 13	< 15
Saline	> 4.0	7.1-8.5	< 13	< 15
Sodic	< 4.0	> 8.5	> 13	> 15
Saline-Sodic	> 4.0	> 8.5	> 13	> 15

Perhaps you are wondering how salted soils develop. There are many sources of salts. These include weathering of primary minerals, organic matter decomposition or addition as fertilizer compounds. Salted soils also develop where seepage water does not drain but evaporate or where irrigation is done with minimal water with no leaching of salts. High concentrations of salts in irrigation water may also be a cause of salinity and sodicity.

6.5.3.2.2 Effects of soil salinity and sodicity on soil properties

Sodic soils make the soil basic with pH values greater than 8.5. Additionally, high levels of carbonates raise the soil pH to levels which may induce deficiencies of micronutrients such as iron and zinc.

Sodium salts cause deflocculation or dispersion of soil particles which results in poor soil structure. The soils become impermeable or less permeable to water due to sealing of pores by entrapped small soil particles dispersed by sodium. These soils keep a layer of water on the surface due to low water infiltration. This type of soil is difficult to till due to formation of hard surface crusts when dry. Soil crusting may inhibit seedling emergence.

High salt contents in soils increase the potential forces that hold water in soils which makes it difficult for plants to extract soil moisture. During dry periods, the high salt concentration can result in death of plants by pulling water from them through exomosis (osmosis in which water flows from a cell or organism into the surrounding solution).

High levels of some salts are toxic to plants. For example, sodium or chlorides in leaves may cause leaf damage. In salty soils, calcium often forms insoluble carbonates or sulphates and

this reduces the availability of calcium for plant uptake. This may also cause nutrient imbalances in plants.

6.5.3.2.3 Management of soil salinity and sodicity

Removal of salts is by leaching using irrigation water which washes away the dissolved salts from the soil profile. Sufficient water should be applied so as leach the salts in drainage water. The soil to be leached should have a high hydraulic conductivity (ability of the soil to transmit water) and adequate drainage system. Plant species have variable tolerances to the salt in soils. Therefore, farmers can also be advised to grow salt-tolerant crops.

To reclaim sodic soils, amendments containing divalent cations (usually calcium) are applied to the soil. The common practice used is the application of gypsum which reduces the sodium content of sodic soils by replacing sodium ions with calcium through cation exchange reactions.

Reclamation of saline-sodic soils is aimed at removing the excess soluble salt and reducing the exchangeable sodium level to below 15 percent. Washing out salts before reducing the sodium content of soils results in dispersion of particles and this makes the soils impermeable to water. The sodium must first be replaced by another cation before leaching it out of the root zone. Therefore, gypsum must be added to the soil before leaching of salts is initiated. Alternatively,

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